



Survival Analysis of Breast Cancer Recurrence

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Introduction

Research Questions

- How do we predict breast cancer recurrence?
- Which features yield a higher recurrence rate?

Dataset

922 patients with 98 features including Demographic, **Tumor Characteristics**, Image Findings, **Treatment Modalities**, **Response to Therapy** and **Follow-up**.¹

- Inclusion criteria: patients who **underwent surgery**
- Time to event: **surgery to recurrence**

Data Cleaning

Challenges: Identifying important variables despite:

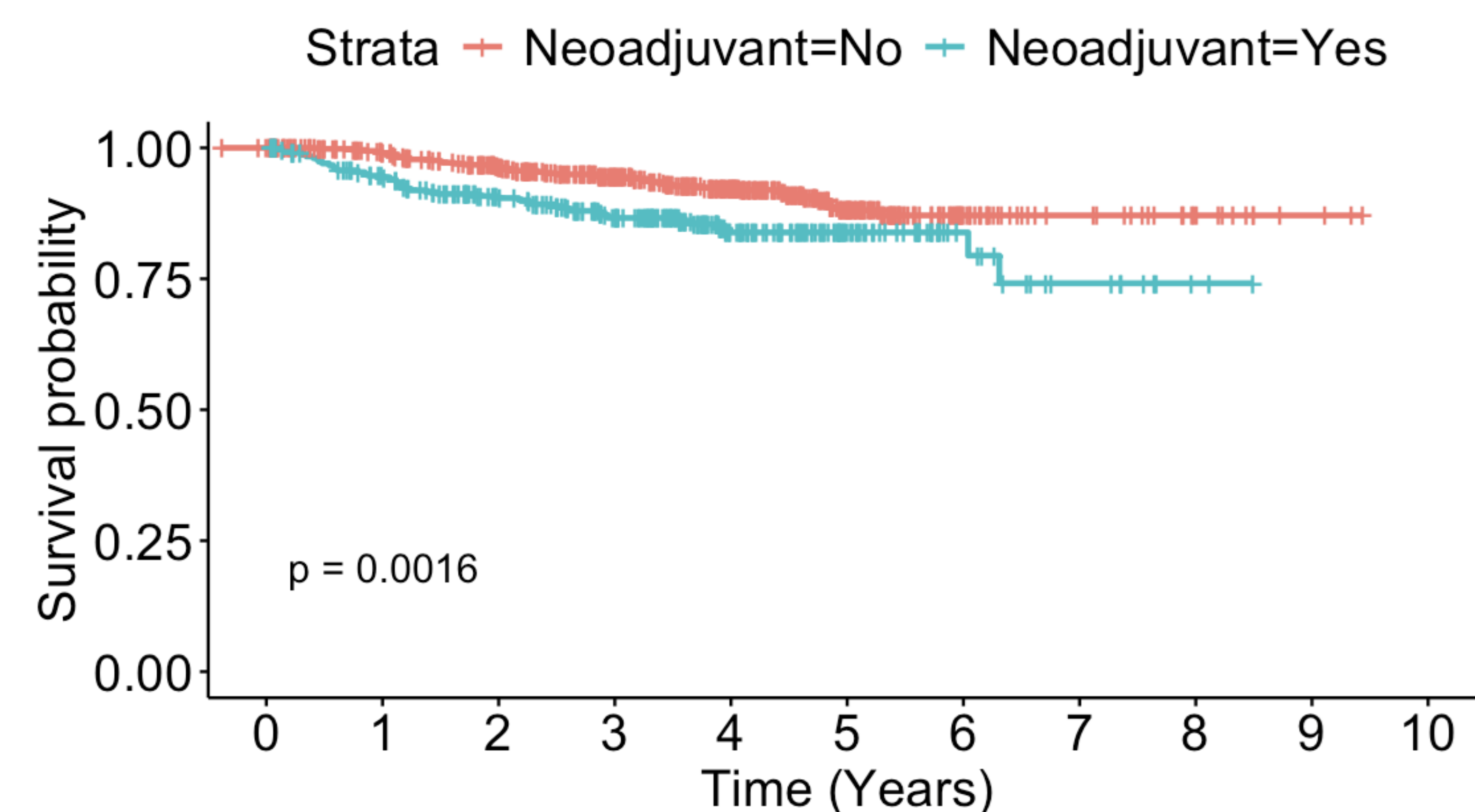
- Sparse observations** for events
- Imbalanced category** distributions
- NOT** randomly missing data (MNAR)

Variable Selection

Remedy: Choosing variables carefully by:

- Relative significance** in naïve log-rank test
- Omitting **largely unobserved** variables
- Consolidating** categorical variables levels

Kaplan-Meier Curve for Neoadjuvant Therapy



Methods: Subgroup Analysis

Interpretable model: Stratified Cox Regression

Semi-parametric Model: **Proportional-Hazard** Assumption

Interpretable Model Results

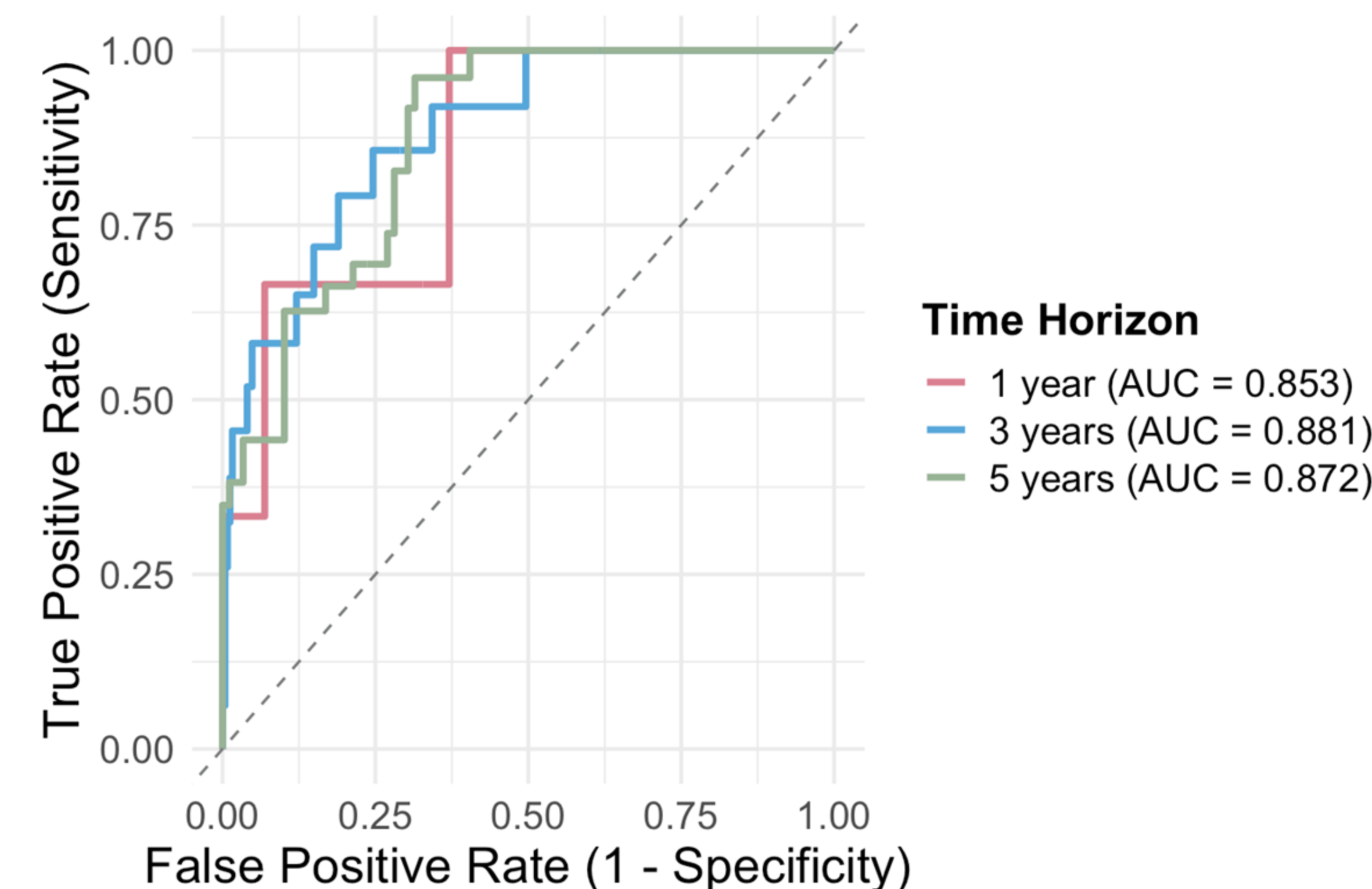
For **luminal A** patients **without prior neoadjuvant** treatment

Cox Regression Summary Table				
Rows with different significant level				
Variable	Reference	Hazard Ratio	95% CI	P-value
Surgery_type = mastectomy	Surgery_type = BCS	4.057	(1.075, 15.317)	0.039
Time_to_surgery	Continuous	1.008	(1.003, 1.013)	0.001
Nottingham_score	Continuous	1.559	(1.059, 2.296)	0.024
N_stage = 1	N_stage = 0	1.084	(0.304, 3.866)	0.901
N_stage = 2	N_stage = 0	6.607	(1.326, 32.935)	0.021
N_stage = 3	N_stage = 0	1.034	(0.088, 12.121)	0.979
T_stage = 2	T_stage = 1	2.897	(1.013, 8.287)	0.047
T_stage = 3	T_stage = 1	4.263	(0.748, 24.311)	0.103
Chemo = None	Chemo = Adjuvant Only	2.294	(0.779, 6.752)	0.132
Radiation = None	Radiation = Adjuvant Only	1.493	(0.423, 5.272)	0.534
Endocrine = None	Endocrine = Adjuvant Only	4.561	(1.382, 15.050)	0.013

Interpretations:

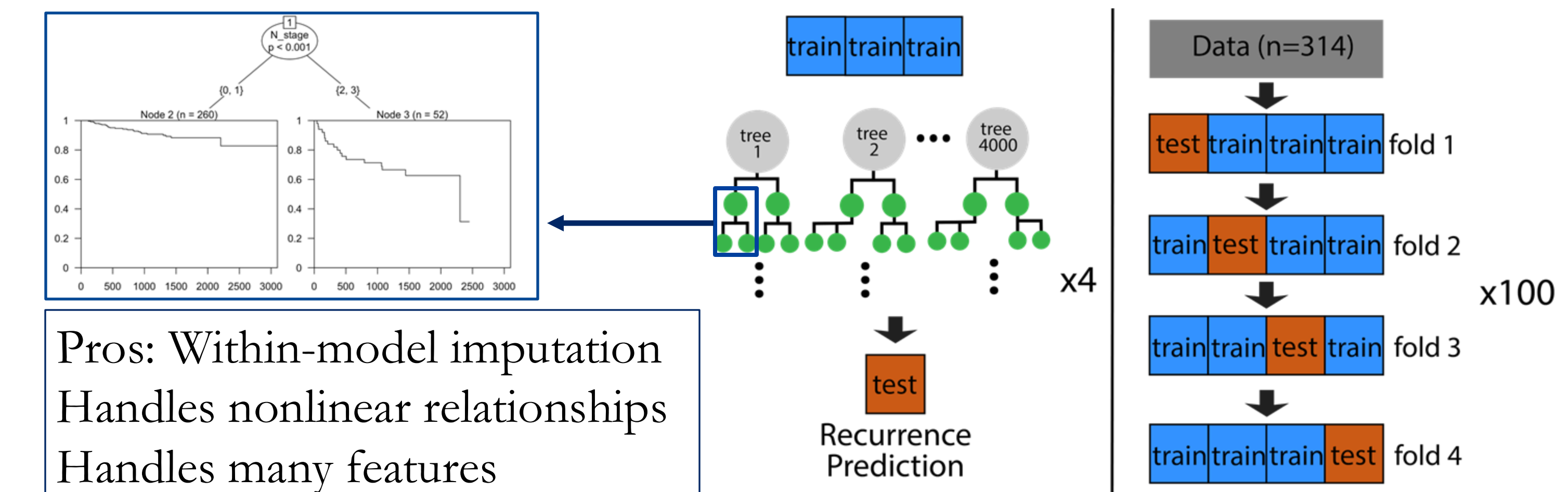
- Patients without endocrine therapy have a **fourfold** higher risk of recurrence compared to those who do
- Delaying surgery by **one month** after diagnosis is associated with **25%** increased hazard of recurrence
- Each **two-point increase** in Nottingham score would **double** the risk of recurrence

Time-dependent ROC Curve with AUC



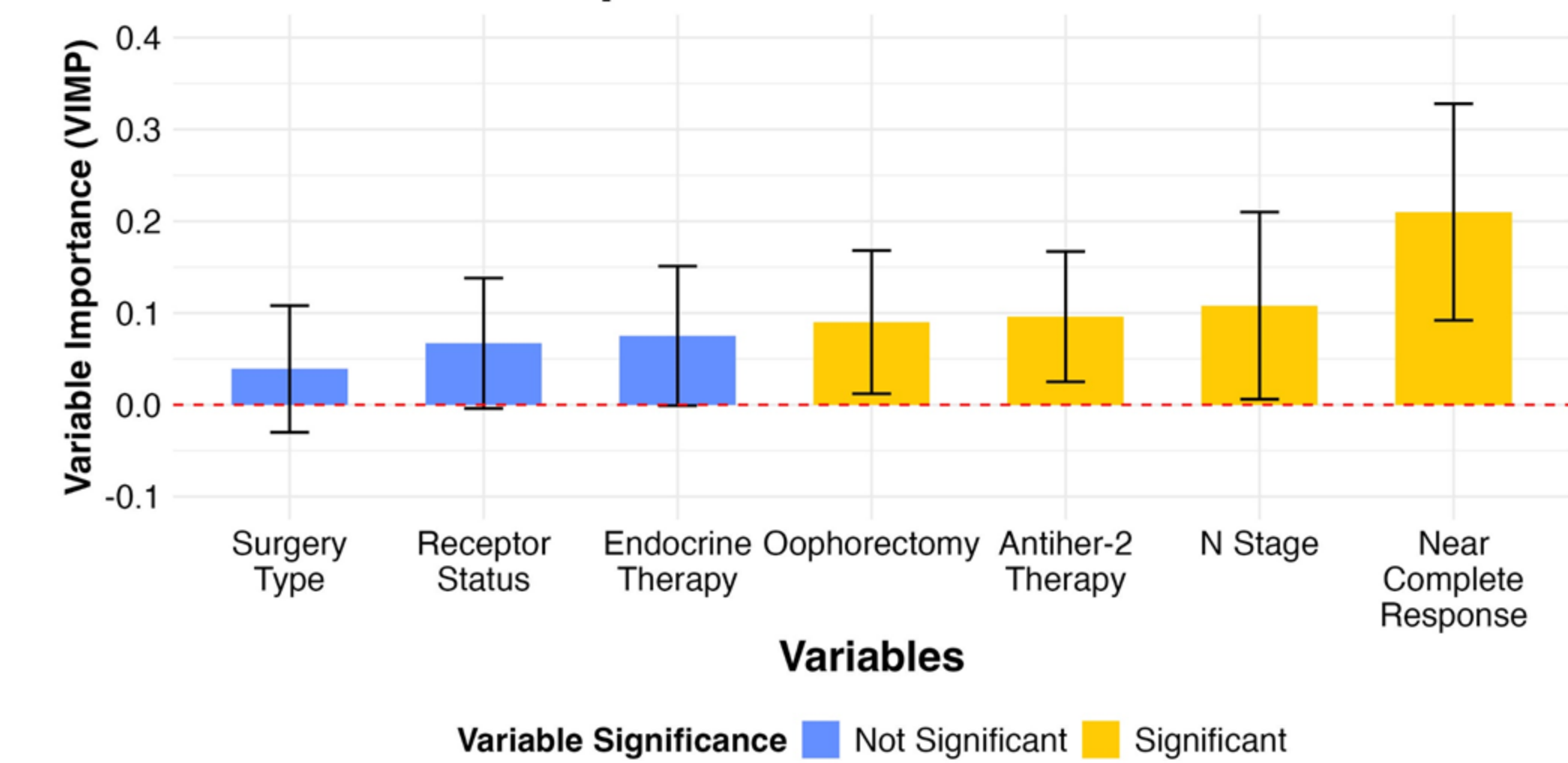
Predictive Model Methods

Repeated K-fold Survival Forest



Predictive Model Results

The Importance of Variables Used

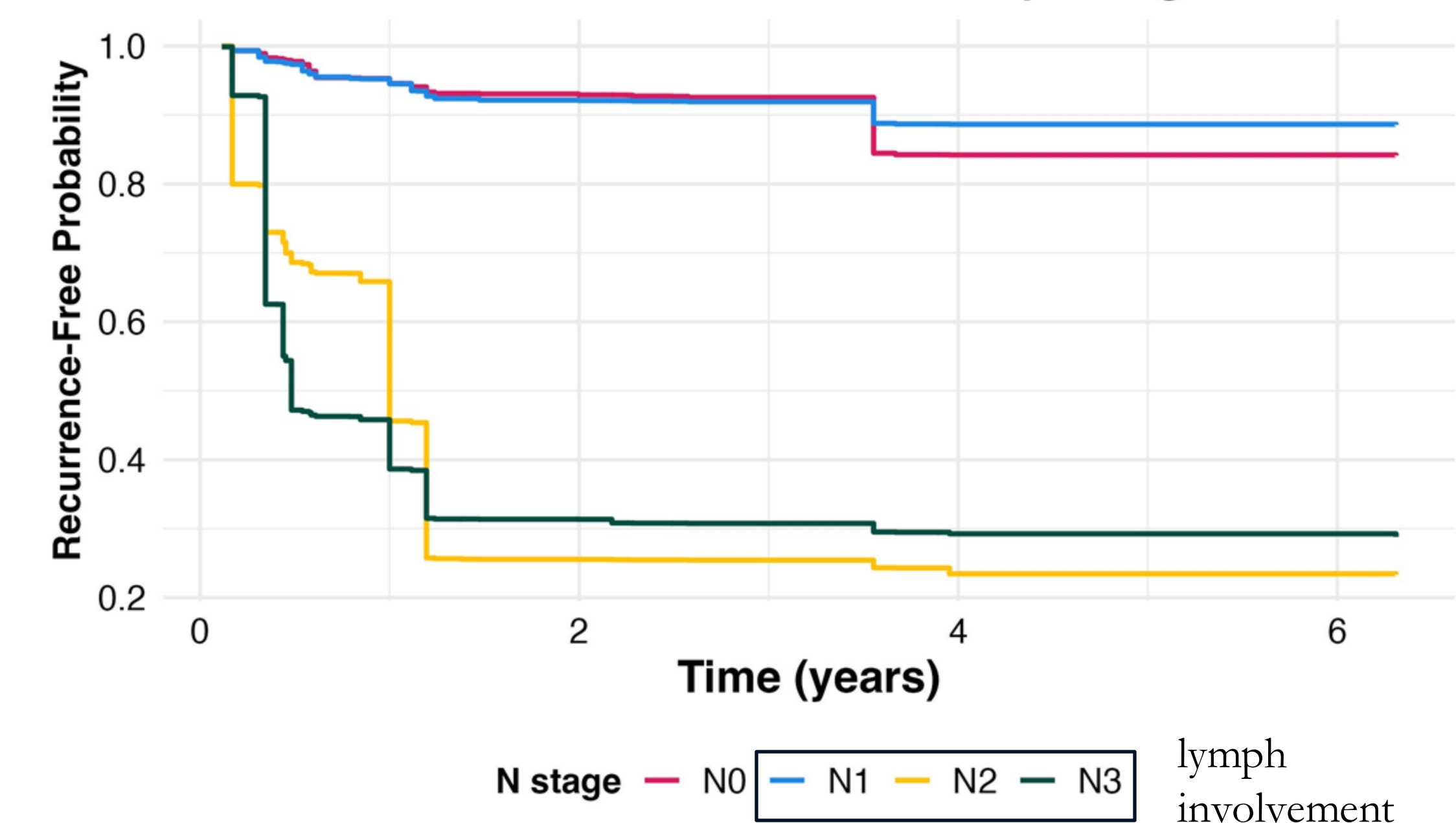


- VIMP scores and standard errors of the variables
- No variables significantly harm performance.
- C-index: .775** [.771, .780].

Clinical Application of Predicting Recurrence

Receptor Status	HER2 therapy	Endocrine Therapy	Node (N) Stage	Surgery Type	Oophorectomy	Near Complete Response
Triple Negative	None	None	Vary	BCS	None	Incomplete

Predicted Recurrence-Free Curve for Triple Negative Cancer



- Recurrence-free survival curves for the patient by N stage
- Relationship between N stage and recurrence probability.
- Grouping in node involvement

Conclusions

- Cox regression offers clinically **interpretable insights**
 - Random survival forests achieve **robust** and **high** discrimination
- ### Future Directions
- Applying similar method to more meaningful subgroups
 - Utilizing other machine learning methods

References

1. Saha, A., Harowicz, M.R., Grimm, L.J., Kim, C.E., Ghatge, S.V., Walsh, R. and Mazurowski, M.A., 2018. A machine learning approach to radiogenomics of breast cancer: a study of 922 subjects and 529 DCE-MRI features. British journal of cancer, 119(4), pp.508-516.